

Early detection of crack vulnerability in foldable displays through critical angle curvature analysis

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Abstract

Foldable displays have become integral to consumer electronics, yet they remain susceptible to mechanical failures, particularly cracks in the hinge region caused by repeated mechanical stress. This study introduces an artificial intelligence (AI)-based method to predict and prevent crack formation by analyzing hinge surface curvature measurements captured within 0.2 seconds of unfolding to 160°, at critical vulnerable angles, using Principal Component Analysis (PCA) and uncertainty quantification with k-Nearest Neighbors (k-NN). A mathematical model incorporating exponential distance-based weighting quantified classification uncertainties, distinguishing confidently classified samples from uncertain cases. Furthermore, an AI-driven scoring model was validated through leave-one-out cross-validation (LOOCV) on an expanded dataset of 200 samples. This model successfully translates complex curvature data into numerical scores, achieving an F1 score of 0.9692 under conditions ensuring zero false positives, thus preventing defective products from reaching customers. This approach significantly enhances quality control in foldable display manufacturing.

KEYWORDS

crack prediction, curvature measurement, foldable displays, hinge mechanism

1 | INTRODUCTION

Foldable displays have emerged as a groundbreaking innovation in consumer electronics, offering a seamless transition between compact and expansive screen functionality. However, their reliability remains a major concern due to inherent mechanical stress during repeated bending cycles, which can lead to cracks, delamination, and other structural failures.¹ The critical nature of these challenges has prompted extensive research into the mechanisms underlying failures and predictive methods for crack detection and prevention.

Foldable displays typically face four primary failure mechanisms: (1) plastic deformation from repeated

strain, (2) fracture in brittle layers such as cover glass, (3) delamination due to interlayer stresses, and (4) buckling-related defects such as warping or wrinkling.² Designers strategically position fragile layers near the neutral bending plane, where mechanical strain is minimal, to mitigate these issues.³ Furthermore, innovations in hinge and panel structures are being explored to minimize stress concentrations and improve durability.⁴

A significant advancement in durability has involved developing reinforced structures such as patterned thin-film transistor (TFT) arrays, which effectively reduce stress concentrations and prevent crack propagation in foldable OLED panels.⁵ For instance, periodic pattern supporting layers have demonstrated their capability to

reduce modulus in the bending zone, thereby enhancing the flexibility of OLED panels and significantly mitigating cracking and peeling problems.⁶ This approach has shown promising results in finite element simulations, highlighting the critical role of optimized structural designs in maintaining display reliability under bending conditions. Another approach optimizes cover-window materials using siloxane-based hybrid polymer coatings, employing two-point bending tests and Weibull statistical models for reliability prediction under varying curvature conditions.⁷

Early and reliable crack detection is essential for ensuring the longevity of foldable displays. Multiple non-destructive testing methods have been developed, including flexible electronic sensors, ultrasonic guided waves, and eddy-current sensing.⁸ Recent studies have shown flexible electronic materials themselves can function as embedded sensors, detecting cracks through measurable variations in electrical resistance.⁹ Furthermore, principal component analysis (PCA) has been successfully utilized in structural health monitoring to identify fatigue-induced cracks and structural anomalies from high-dimensional sensor data.¹⁰ High-resolution image-based techniques, such as robust principal component analysis (RPCA), further enhance crack detection by quantifying surface defects under stress.¹¹

AI-driven inspection systems have increasingly become central in defect detection in display manufacturing. Deep learning techniques, particularly convolutional neural networks (CNNs), provide superior accuracy compared to traditional image processing methods, facilitating real-time defect identification in various industrial applications.¹² Additionally, statistical and machine learning approaches such as PCA and k-Nearest Neighbors (kNN) algorithms offer robust frameworks for uncertainty quantification and early failure prediction by analyzing bending strain distributions during repeated folding cycles.⁷

Inspired by recent advancements in machine learning applications in biomedical diagnostics—such as differentiating diabetic from healthy samples using fluorescence spectroscopy with random forest and KNN algorithms,¹³ LIBS spectra of fingernails analyzed using PCA and machine learning classifiers,^{14,15} and Raman spectroscopy coupled with SVM models¹⁶—similar approaches leveraging PCA, random forests, and support vector machines (SVMs) hold potential in foldable electronics for accurately diagnosing early-stage cracks or stress-induced material degradation.

As foldable displays become increasingly prevalent, ongoing efforts to understand and address reliability issues through material innovations, structural design enhancements, advanced sensor integration, and data-

driven predictive modeling are crucial. Such multidisciplinary strategies promise substantial improvements in the reliability and commercial viability of foldable display technologies.

As foldable and flexible displays become more prevalent, ensuring their mechanical reliability is a critical area of research. Current studies address crack formation and propagation through material innovations, predictive modeling, and AI-enhanced defect detection. By integrating advanced sensing, machine learning, and statistical modeling, researchers aim to develop smart monitoring systems that can predict and prevent failures before they compromise device integrity. Continued research into multi-modal defect detection and real-time failure prediction will be essential for the widespread adoption of foldable display technology.

The vulnerability to future cracking is particularly acute in the hinge area, the central folding mechanism of the device, where mechanical stresses are most concentrated during operation. Even minor manufacturing tolerances or defects can become significant over time, causing cracks to initiate and spread after prolonged use (Figure 1). These defects not only compromise the structural integrity and longevity of the device but also reduce user satisfaction and confidence in technology.

In this paper, we propose a novel method leveraging artificial intelligence to analyze and preemptively identify samples that are vulnerable to future crack formation. By utilizing advanced AI algorithms and machine learning techniques, we aim to detect subtle indicators of potential mechanical failure that may not be apparent through conventional inspection methods. This proactive approach allows for the early identification of at-risk components, enabling manufacturers to address issues before they affect product performance.

2 | EXPERIMENTS

2.1 | Hinge curvature PCA analysis

In this study, we investigated the surface curvature of foldable display panels by reflecting a linear light pattern onto them. Preliminary in-house experiments revealed that maximum stress does not occur at full extension (180°) but is instead concentrated near an unfolding angle of approximately 160°. Accordingly, all measurements were taken at this critical angle. Moreover, due to viscoelastic relaxation effects, the curvature undergoes significant changes within several seconds after unfolding; therefore, the curvature measured within 0.2 seconds of actuation was chosen as the key performance indicator (Figure 2).

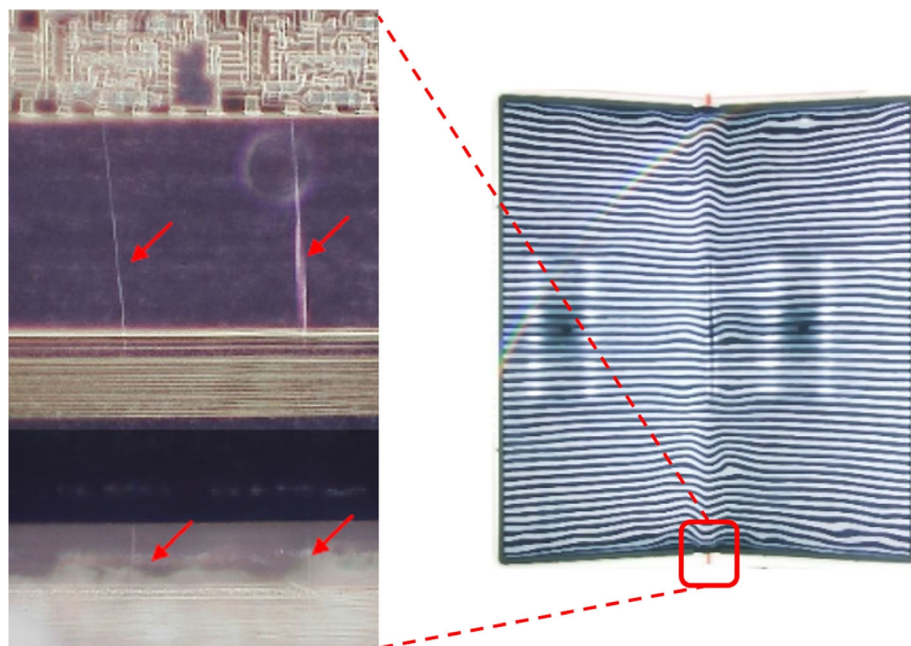


FIGURE 1 This illustrates the development of cracks in the hinge area of a foldable display due to stress accumulation. This region is particularly susceptible to mechanical failure, and prolonged cycles of opening and closing exacerbate stress, leading to crack initiation and propagation as shown in the left image.

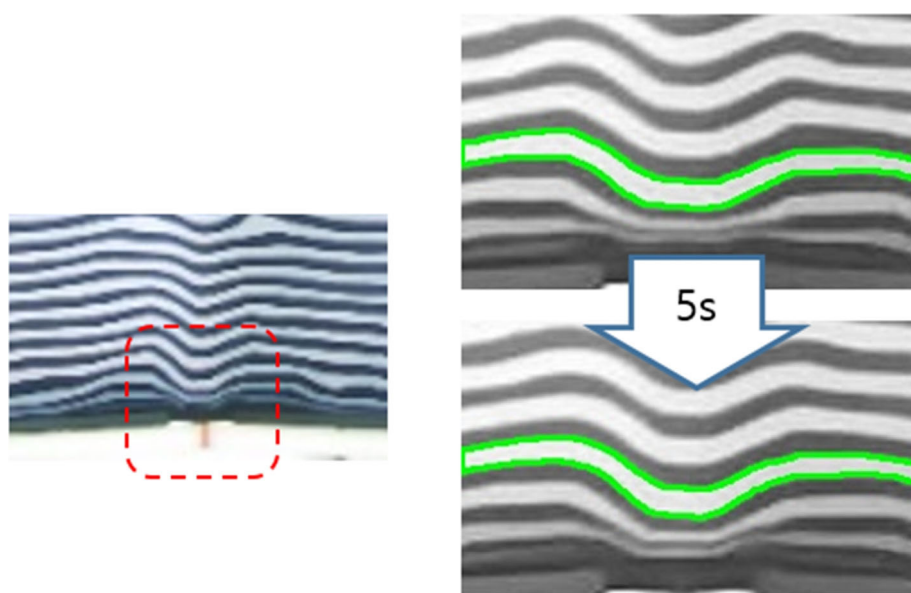


FIGURE 2 This illustrates the importance of initial measurement when assessing the curvature of a foldable display's surface at a specific angle (160 degrees). Since the device utilizes flexible display elements, its surface curvature can change over time. Therefore, early measurements are crucial to accurately capture the device's initial bending characteristics.

Through rigorous stress testing involving repeated cycles of opening and closing, we collected data on both defective (cracked) and non-defective samples, specifically measuring the initial curvature at the critical vulnerable angle. These curvature profiles were subsequently subjected to Principal Component Analysis (PCA),¹⁷ reducing the high-dimensional data to two principal components that account for approximately 60% of the total variance (Figure 3). Since the curvature image data are inherently high-dimensional and the sample size is relatively limited, direct visualization of the original data is challenging. Although incorporating principal components PC3 through PC5 could extend the explained

variance to approximately 85%, we opted to use only PC1 and PC2 for visualization in a two-dimensional space to enhance interpretability.

In the resulting two-dimensional PCA plot, samples that developed cracks are represented by red dots, while non-defective samples are depicted as blue dots. Despite the limited number of samples—due to constraints inherent in reliability evaluations—the scatter plot reveals a general tendency and suggests a potential boundary between defective and non-defective samples. However, a clear and definitive separation is not evident.

This indistinct demarcation may be attributed to several factors. First, utilizing only two principal

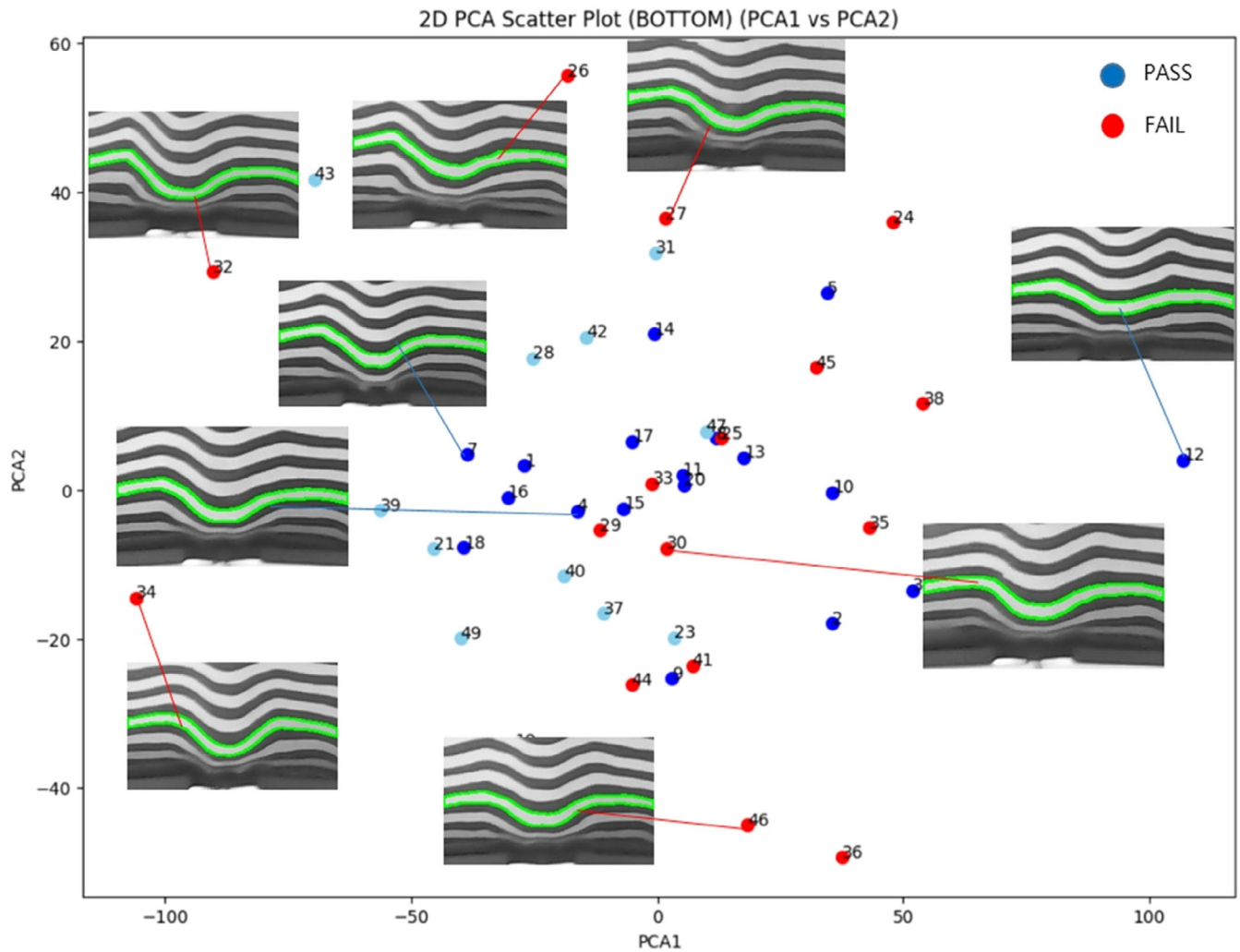


FIGURE 3 This figure illustrates the two-dimensional distribution obtained by applying principal component analysis (PCA) to the bending shapes of the hinge area in foldable devices measured at the critical angle. Blue dots represent normal samples, while red dots indicate samples where crack defects have occurred. This visualization highlights the differences in bending characteristics between normal and defective samples.

components means that not all of the original data's information is captured, potentially omitting critical features necessary for accurate classification. The retained 60% of variance might be insufficient to fully represent the complex curvature profiles associated with crack formation. Second, relying solely on the initial curvature measurements at the vulnerable angle may not provide adequate discriminatory power to distinguish between samples that will eventually develop cracks and those that will not.

To further investigate these limitations and enhance the predictive capability of our analysis, it is imperative to incorporate additional uncertainty analysis. By quantifying the uncertainty associated with each sample's position in the PCA space, we can gain a deeper understanding of the overlap between defective and non-defective samples. This approach may help determine whether the ambiguity arises from intrinsic data

variability, measurement errors, or inadequate feature representation. Ultimately, incorporating uncertainty measures could facilitate the development of more robust classification boundaries and improve the overall reliability of crack prediction in foldable displays.

2.2 | Uncertainty analysis and modeling

In this study, we aim to analyze and quantify the uncertainty inherent in the curvature measurements and reliability evaluation results obtained from our experiments. By addressing this uncertainty, we seek to more precisely identify and distinguish between samples prone to crack formation and those considered safe.

To achieve this, we developed a mathematical model to calculate an uncertainty score for each data point

based on its proximity to neighboring samples and their respective class labels (defective or non-defective). This approach is based on the k -nearest neighbors (k -NN) algorithm, a widely used nonparametric classification method first introduced by Fix and Hodges¹⁸ and further formalized by Cover and Hart.¹⁹ Specifically, k -NN provides a probabilistic interpretation of classification uncertainty by examining the class distribution among nearest neighbors, making it particularly effective for identifying borderline cases or samples with ambiguous class memberships.^{20,21}

The uncertainty score for a given data point is defined using the k -nearest neighbors approach as shown in equation (1):

$$U(x_i) = \frac{\sum_{x_j \in N_k(x_i), \text{Class}(x_j) \neq \text{Class}(x_i)} w(d_{ij})}{\sum_{x_j \in N_k(x_i)} w(d_{ij})} \quad (1)$$

where:

x_i is the data point for which we are calculating the uncertainty.

$\text{Class}(x_i)$ denotes the class label of x_i (either defective or non-defective).

$N_k(x_i)$ represents the set of k -nearest neighbors of x_i with distances d_{ij} for each neighbor x_j in $N_k(x_i)$.

$w(d_{ij}) = e^{-d_{ij}}$ is a weighting function based on distance, assigning higher weights to closer neighbors.

The exponential weighting function $e^{-d_{ij}}$ was specifically chosen for its balanced properties, providing smooth, continuous weighting without the instability or singularities that can arise from alternative weighting functions such as inverse distance weighting or linear interpolation methods.^{20–22} The exponential form ensures that closer neighbors have significantly greater influence

while distant neighbors contribute minimally, thus effectively localizing the uncertainty estimation.

This approach aligns well with recent developments where distance-weighted k -NN classifiers have demonstrated improved performance in terms of stability, accuracy, and robustness, especially in scenarios involving complex data distributions and ambiguous classification boundaries.^{23–25} By leveraging these theoretical insights, our proposed uncertainty model robustly quantifies the reliability of classification, clearly distinguishing between confidently classified samples and those requiring further scrutiny.

By incorporating these theoretical principles, our uncertainty score not only quantifies the reliability of the classification but also provides a clear distinction between samples that are confidently classified and those that lie near the decision boundary, requiring further attention.

The numerator of the equation sums the weighted contributions of neighboring samples whose class labels differ from that of, while the denominator sums the weighted contributions of all neighboring samples. An uncertainty score close to 1 indicates high uncertainty (as most nearby samples belong to a different class), whereas a score close to 0 indicates low uncertainty (as most nearby samples share the same class) (Figure 4).

By applying this method, we can effectively mitigate the uncertainty arising from experimental measurements and enhance the precision of our analysis. This allows us to more accurately quantify and distinguish between samples that are susceptible to crack formation and those that are not.

The results of this uncertainty analysis are depicted in Figure 5. The curvature shapes measured at the critical vulnerable angle reveal distinct patterns. Samples that are safe exhibit a sharp V-shaped curvature, indicating a

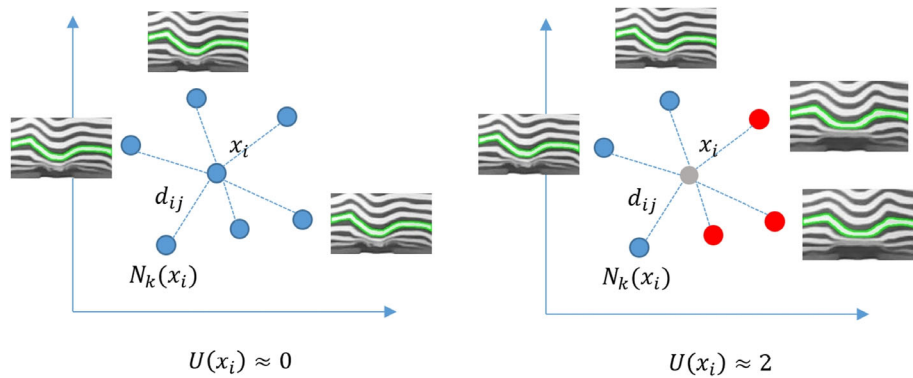


FIGURE 4 This illustrates the concept of measuring the uncertainty of information obtained from experiments and measurements, which is crucial during the research stage when the number of samples is limited. In the PCA plot, data points surrounded by homogeneous information (i.e., similar nearby data points) are considered to have higher confidence. Conversely, data points with a heterogeneous neighborhood are judged to be more uncertain. This visualization emphasizes how the proximity of similar data in PCA space can influence the assessment of data reliability under conditions of sample scarcity.

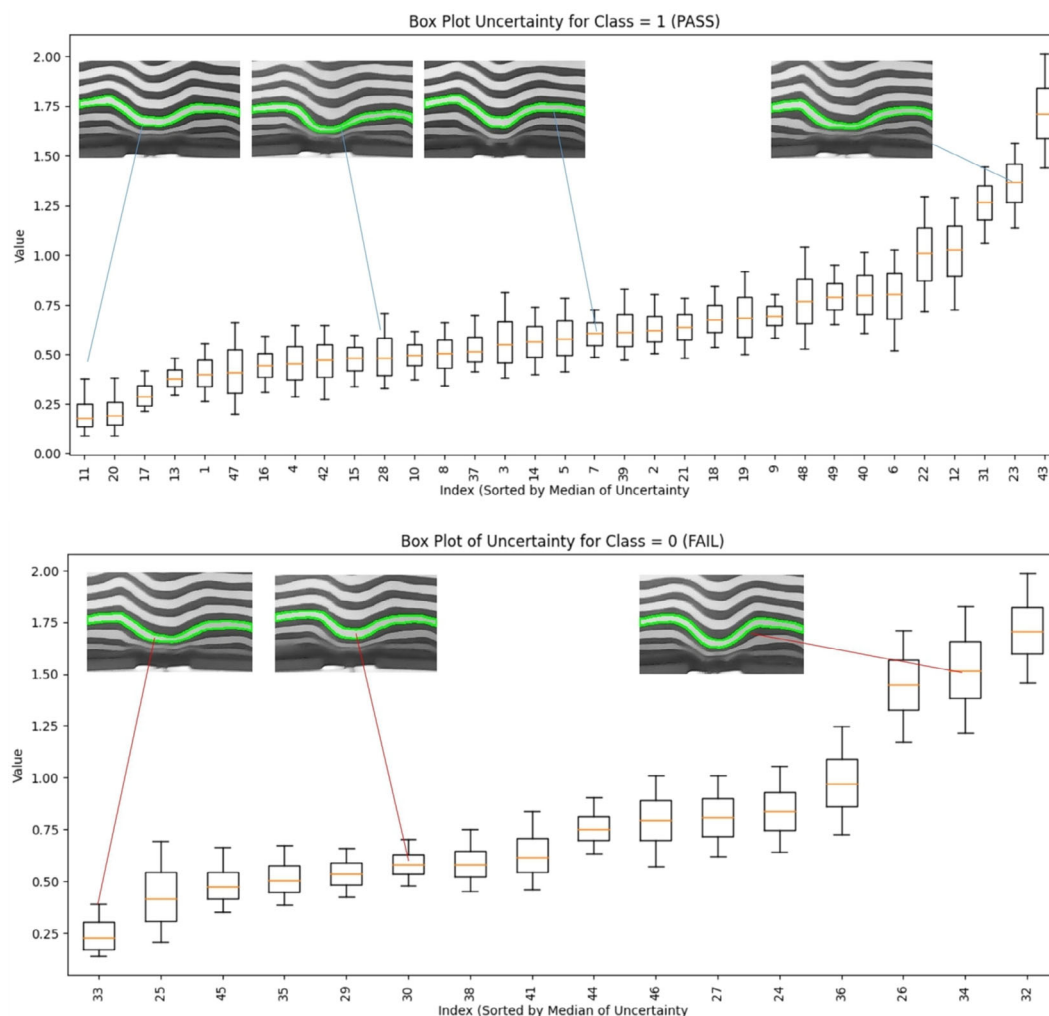


FIGURE 5 This presents the results of measuring the uncertainty for each sample. Since the uncertainty values vary depending on the number of neighboring samples referenced, we have depicted their distribution. The upper left image clearly demonstrates a strong tendency toward being highly resistant to cracks. In contrast, although the sample in the upper right image passed the reliability evaluation, it exhibited a vulnerable form. Additionally, the lower left image shows a tendency toward being highly susceptible to crack occurrence.

more acute bend. In contrast, samples prone to cracks display a slightly flattened U-shaped curvature. This difference suggests that deviations from the original design specifications—possibly due to minor production errors—can lead to variations in stress distribution during folding operations.

The presence of these variations implies that certain samples may experience abrupt changes in mechanical stress, increasing their susceptibility to crack formation over time. By employing our uncertainty analysis method in conjunction with AI techniques, we can detect these subtle differences in curvature profiles. This enables us to preemptively identify and classify at-risk samples during the production process.

Building upon the uncertainty measurements derived from our initial sample set, we developed an AI-driven scoring model designed to assess each product's risk of

future crack formation. This model assigns each sample a numerical score ranging from +1 to −1, where +1 represents a clearly non-defective (good) product, and −1 signifies a product highly susceptible to cracking. To validate the effectiveness of this model, we applied it to an expanded dataset comprising 200 samples, with the evaluation results presented in Figure 6.

Our methodology utilized a K-Nearest Neighbors (KNN) classifier combined with leave-one-out cross-validation (LOOCV), an approach particularly suitable for small datasets.^{26,27} LOOCV effectively maximizes data utilization by systematically treating each data point as a test sample exactly once, ensuring optimal use of available training data.²⁸ This technique significantly reduces potential biases introduced by arbitrary data partitioning, resulting in a more stable and dependable model evaluation.²⁹ Additionally, LOOCV mitigates overfitting risks,

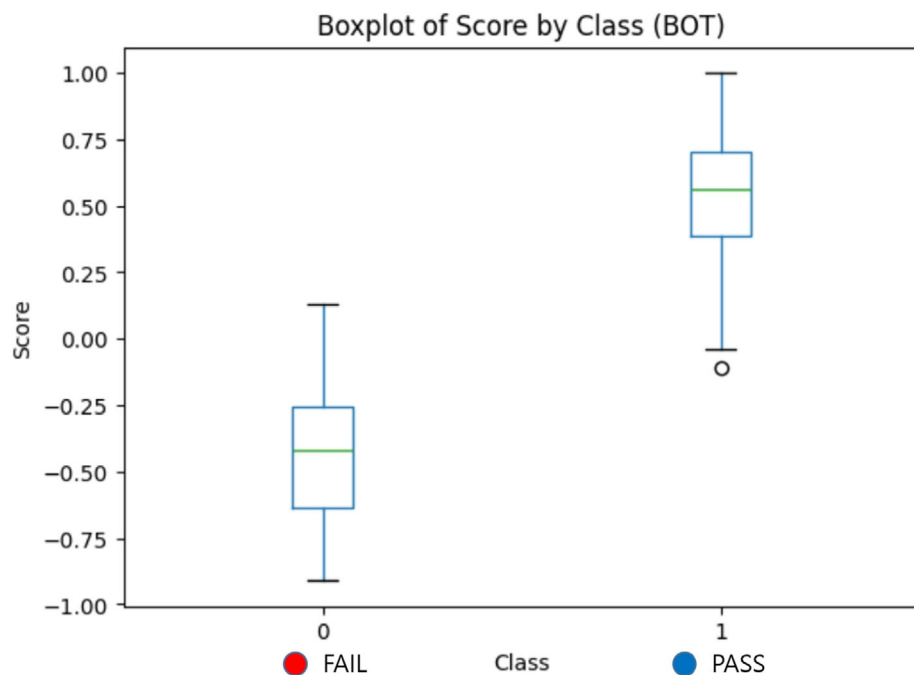


FIGURE 6 This presents the results of measuring a large volume of mass production data after modeling based on the previously assessed judgment uncertainty. The data have been recalculated into a score value, where a score of +1 indicates a good product with strong confidence, and values approaching -1 signify a defective product with strong confidence. Around the score of 0—an interval where judgment is difficult—there is an overlap between good and defective products. However, to minimize leakage, the decision boundary can be strategically restructured. This allows for the adjustment of classification thresholds to better separate good products from defective ones, enhancing the reliability of the assessment.

enhancing the generalizability and robustness of our findings.³⁰

The classifier's performance was evaluated by identifying a critical threshold score of 0.1303, which ensures no defective products would erroneously pass through quality inspections (False Positive rate = 0). At this threshold, the classifier exhibited outstanding performance metrics, achieving a precision of 1.0, an accuracy (Acc) of 0.9572, an F1 Score of 0.9692, and a recall of 0.9403.

Overall, this AI-driven scoring system provides an innovative method to quantify crack risk—something previously unattainable through conventional analytical approaches. By translating complex curvature and stress data into a straightforward numerical score, our model significantly improves both the precision and practical applicability of product quality assessments, thereby offering substantial enhancements to quality control and customer satisfaction in manufacturing environments.

3 | CONCLUSION

In this study, we conducted a comprehensive investigation into the curvature characteristics of foldable display panels, focusing on critical angles and initial bending measurements to effectively identify products susceptible to crack formation. Utilizing PCA, we successfully visualized complex, high-dimensional curvature data, uncovering general trends between defective and non-defective samples, although a distinct separation remained elusive.

Recognizing these limitations, we implemented an uncertainty quantification methodology based on a k-Nearest Neighbors (k-NN) approach combined with exponential distance-based weighting. This method effectively quantified the ambiguity inherent in classifying samples, distinguishing clearly confident classifications from those situated near uncertain boundaries.

The uncertainty analysis notably enhanced our ability to detect subtle differences in curvature profiles, enabling early identification of samples potentially prone to cracking due to minor manufacturing variations. Building upon these uncertainty measurements, we developed an AI-driven scoring model leveraging k-NN classification and leave-one-out cross-validation (LOOCV). This innovative approach translated intricate curvature and stress data into intuitive numerical scores, significantly enhancing prediction reliability.

Validation using an expanded dataset of 200 samples demonstrated exceptional model performance, achieving a precision of 1.0, accuracy of 95.72%, an F1 Score of 96.92%, and a recall of 94.03%, with no false positives identified at the defined critical threshold. Thus, our proposed methodology not only addresses critical gaps in conventional analytical approaches but also substantially advances practical quality assessment processes in foldable display manufacturing. Ultimately, this research provides manufacturers with robust tools for improving product reliability, mitigating risks, and enhancing customer satisfaction. For future research, we propose exploring a hybrid approach combining physical modeling of stress distribution in hinge areas with data-driven

AI models. This will further enhance the robustness and extend the applicability of our methodology, ultimately contributing to a deeper understanding and improved predictive capability for foldable display reliability.

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AUTHOR BIOGRAPHIES



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